

JEE MAIN REVISION MASTER: GEOMETRICAL OPTICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 268 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Refraction at Interfaces, Snell's Law & TIR	143 PYQs	Refraction
Very High	Reflection at Spherical Mirrors & Mirror Formula	127 PYQs	Mirrors
Very High	Lens Maker's Formula & Optical Instruments	97 PYQs	Lenses & Instruments
Very High	Prism Dispersion & Minimum Deviation	48 PYQs	Prisms

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Refraction at Interfaces, Snell's Law & TIR	$n_1 \sin i = n_2 \sin r$, $\sin \theta_c = \frac{n_2}{n_1}$; ($n_1 > n_2$), $d' = \frac{d}{\mu}$, $\text{Shift} = t \left(1 - \frac{1}{\mu} \right)$
Reflection at Spherical Mirrors & Mirror Formula	$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, $m = -\frac{v}{u} = \frac{f}{f - u}$, $f = \frac{R}{2}$
Lens Maker's Formula & Optical Instruments	$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$, $P = \frac{1}{f}$, $M_{\text{telescope}} = \frac{f_o}{f_e}$, $M_{\text{microscope}} = \frac{L}{f_o} \frac{D}{f_e}$
Prism Dispersion & Minimum Deviation	$\delta = i + e - A$, $\mu = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin(A/2)}$, $\delta \approx (\mu - 1)A$; (thin prism)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Refraction at Interfaces, Snell's Law & TIR (Very High Yield — 143 PYQs)

Core Principle: Snell's law, apparent depth shift in transparent media, and total internal reflection condition ($i \geq \theta_c$ from denser to rarer).

■ **Caution / JEE Pitfall:** Total internal reflection can ONLY occur when light travels from an optically DENSER medium to a RARER medium.

■ **Speed Trick:** Area of circle through which light emerges from fish/source at depth h : $A = (\pi h^2 / \mu^2 - 1)$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A po..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-con..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be " [Ans: B]

3.2 Reflection at Spherical Mirrors & Mirror Formula (Very High Yield — 127 PYQs)

Core Principle: Image formation by concave and convex mirrors, Cartesian sign convention, and linear/longitudinal magnification.

■ **Caution / JEE Pitfall:** Concave mirror has negative focal length ($f < 0$); convex mirror has positive focal length ($f > 0$).

■ **Speed Trick:** Virtual image in a concave mirror is ALWAYS magnified ($|m| > 1$), while in a convex mirror it is ALWAYS diminished ($|m| < 1$).

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A po..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-con..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a plano-concave lens (refractive index =1.7) are same. If t..." [Ans: B]

3.3 Lens Maker's Formula & Optical Instruments (Very High Yield — 97 PYQs)

Core Principle: Thin lens refraction, Lens Maker's equation, lens power in Dioptres ($P = 1/f$), and magnifying power of optical devices.

■ **Caution / JEE Pitfall:** When a lens is immersed in a liquid of refractive index $\mu_l > \mu_{\text{(glass)}}$, its converging/diverging nature REVERSES (convex acts concave).

■ **Speed Trick:** When a lens is cut vertically in half, each half retains the SAME focal length f . When cut horizontally, focal length remains f , but image brightness is halved.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-con..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a plano-concave lens (refractive index =1.7) are same. If t..." [Ans: B]
- [2026 Evening Shift] Q#1 (Numerical): "The size of the images of an object, formed by a thin lens are equal when the object is placed at two different positions 8 cm and 24 cm fro..." [Ans: 16]

3.4 Prism Dispersion & Minimum Deviation (Very High Yield — 48 PYQs)

Core Principle: Ray refraction through triangular prisms, condition for minimum deviation ($i = e$, $r_1 = r_2 = A/2$), and angular dispersion.

■ **Caution / JEE Pitfall:** At minimum deviation, the refracted ray inside an equilateral prism passes strictly PARALLEL to the base.

■ **Speed Trick:** Condition for no emergent ray (always TIR on second face): $A > 2\theta_c$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be _____." [Ans: B]
- **[2026 Evening Shift] Q#1 (MCQ):** "A prism of angle 75° and refractive index $\sqrt{3}$ is coated with thin film of refractive index 1.5 only at the back exit surface..." [Ans: N/A]
- **[2026 Morning Shift] Q#3 (MCQ):** "A thin prism with angle 5° of refractive index 1.72 is combined with another prism of refractive index 1.9 to produce dispersion wi..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of $4R$ in front of spherical interface. The magnitude of the magnification of point source image is _____.

- (A) 1.66 [Correct]
- (B) 2.33
- (C) 2.66
- (D) 1.33

Correct Answer: A

Solution Analysis: When light travels from one medium to another through a curved surface, the positions of the object (u) and the image (v) are related by the formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ Where n_1 is the refractive index of the medium where the object is placed, n_2 is the refractive index of the medium where light enters, R is the radius of curvature. The refractive index of medium in which object is $n_1 = 1$, and that for the image $n_2 = 1.4$. The object distance is $u = -4R$ (as per sign convention), and radius is $+R$ (convex surface). $\frac{1.4}{v} - \frac{1}{-4R} = \frac{1.4 - 1}{R}$

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be _____.

- (A) 3 : 4
- (B) 3 : 2 **[Correct]**
- (C) 2 : 1
- (D) 1 : 2

Correct Answer: B

Solution Analysis: For a thin prism with a small prism angle (A), the angle of minimum deviation is given by, $\delta_{(m)} = (\mu - 1)A$ Where μ is the refractive index of the glass, $\mu = 1.5$: Putting the value of refractive index, $\delta_{(m)} = (1.5 - 1)A = 0.5A = (A / 2)$ At the position of minimum deviation, the angle of incidence (i) is related to the prism angle (A) and the minimum deviation ($\delta_{(m)}$) by $i = \frac{A + \delta_{(m)}}{2}$ Substituting the value of $\delta_{(m)} = (A / 2)$ $i = (A + (A / 2)) / 2 = (3A / 2) / 2 = (3A / 4)$ So, the ratio of the incident angle (i) to the minimum deviation ($\delta_{(m)}$) is $\frac{3A / 4}{A / 2} = \frac{3}{2}$ So, the ratio of the incident angle (i) to the minimum deviation ($\delta_{(m)}$) is 3 : 2.